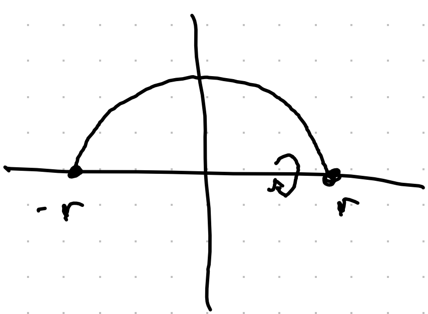
Surface area examples — 2.4

$$SA = \int_a^b 2\pi f(x)\sqrt{1+f'(x)^2}\,dx$$

Example: Find the surface area of a sphere of radius r.



Rotate the curve $\sqrt{r^2-x^2}$

$$f'(x) = \frac{-x}{\sqrt{r^2-x^2}}$$

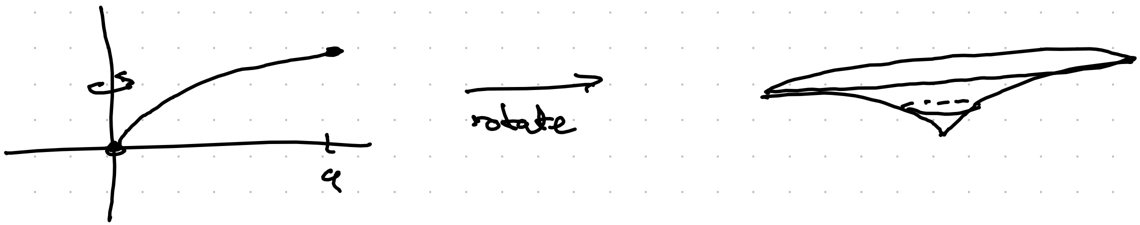
$$SA = \int_{-r}^{r} 2\pi\sqrt{r^2-x^2}\cdot\sqrt{1+\frac{x^2}{r^2-x^2}}\,dx$$

$$= \int_{-r}^{r} 2\pi\sqrt{r^2-x^2+x^2}\,dx = \int_{-r}^{r} 2\pi\sqrt{r^2}\,dx$$

$$= \int_{-r}^{r} 2\pi r\,dx = 2\pi r x\Big|_{-r}^{r}$$

$$= 2\pi r(r-(-r)) = 2\pi r\cdot 2r = 4\pi r^2.$$

Example: Find the area of the surface obtained by rotating $y = \sqrt[3]{3x}$ on $[0,9]$ about the y-axis.



Rotation about y-axis means splitting the y-axis into small pieces, adding them up, and taking a limit to produce an integral.

$\longrightarrow$ need our curve given as a function of the variable y.

Rearranging: $y = \sqrt[3]{3x} \iff y^3 = 3x \iff x = y^3/3$.

$x = 0 \iff y = 0$, $\quad x = 9 \iff y = 3$

$x = y^3/3$ on the interval $[0,3]$

SA after rotation is $\int_0^3 2\pi \cdot \frac{y^3}{3} \sqrt{1+y^4}\, dy$

$= \frac{2\pi}{3} \int_0^3 y^3 \sqrt{1+y^4}\, dy$ $\qquad u = 1+y^4$

$\qquad du = 4y^3\, dy$

$= \frac{2\pi}{3} \cdot \frac{1}{4} \int_1^{82} \sqrt{u}\, du = \frac{\pi}{6} \cdot \frac{2}{3} u^{3/2} \Big|_1^{82}$

$= \frac{\pi}{9} \left(82\sqrt{82} - 1 \right)$